

Data: data is collection of raw facts.

Information: Information is processed data.

What is database?

It is a collection of inter-related data and relationships among those data.

What is database management system?

A database-management system (DBMS) is a collection of inter-related data and a set of programs to access those data.

(or)

A Database Management System (DBMS) is software designed to store, retrieve, define, and manage data in a database.

Popular DBMS Software: Here, is the list of some popular DBMS systems:

- Oracle
- NCR Teradata
- Microsoft SQL Server
- MySQL
- IBM DB2
- Microsoft Access
- SQLite
- MariaDB, etc.

Disadvantages of File System

Generally, we store our data in a text file. Now think about if you want to develop a system software for banking system, then we need to create a lot of text file to store different information. By considering this example we understand disadvantages of conventional based file system.

Disadvantages of File System compare to DBMS

1) Data Redundancy

- Data redundancy means repetition of data.
- There is possibility of data duplication of same information in different files. This situation leads towards data redundancy.
- As a result of data redundancy leads to memory wastage.
- Example: A customer having both kind of accounts- saving and current. In such a situation a customer's detail like: name, address, e-mail, contact number, city, etc... stored in both the files, **saving.txt** - a file which store's details of saving account holder and **current.txt**- a file which store's details of current account holder. because of storing same information of customer in two separate files, it leads to **Data Redundancy**.

2) Data Inconsistency

- Because of data redundancy, may be data is not in a consistent state.
- Example: (Here we continue above example) Suppose a customer having both kind of accounts a saving and a current. so, in file system this customers address is stored in both the files **saving.txt** and **current.txt**. Now customer changed his/her **address**. There might be a possibility that address is changed in only one file (**saving.txt**) and other (**current.txt**) remain unchanged. So now for the same customer in two different file two different value/information is available for the same field/attribute address. This is called **Data Inconsistency**.

3) Difficulty in Accessing Data

- In file processing system Accessing Data is not convenient and efficient.
- Example: We have stored customers data like: name, address, e-mail, contact number, and city. Now we want a complete record of the customer who lives in a particular city (for example, Guntur), to achieve this we need to write a separate program **OR** we need to execute to separate program i) to select all the record from file ii) select only those record whose city is Guntur from the result of first (i) program. We cannot achieve convenient and efficient data accessing by using both ways.

4) Limited Data Sharing/ Data Isolation

- Data are stored in various files.

- File formats may be different (**for example: .txt and .xml**) and these files may be stored in different folders of computer system or may be in a different system.
- Because of isolation of data, it is difficult to share data between various applications.

5) Integrity Problems

- It means the data available in the database is must be correct and consistent.
- To achieve this data stored in a file must satisfy the rules define by the DBA.
- Example: Initially while developing s/w for any bank they have decided that account balance should not be less than zero, and they developed an application program according to this constraint/rule. Now after few years chairperson and committee member of bank have decided that minimum balance should not be less than 2000, to implement these constraints you need to change in application program and need to install in all systems using that application program, but this is not an easy task.

6) Atomicity Problems

- It means, transaction/operation completes either all (100%) or none (0%).
- In other words all (any) operations on database must be atomic.
- Example: Suppose there are two bank accounts: **A/c-1** and **A/c-2** having balance Rs.4500 and Rs.2000 respectively. Now we have to transfer Rs.500 from **A/c1** to **A/c-2**. To achieve this, entire transaction divide into two separate parts: i) debit Rs.500 from **A/c-1** and ii) credit Rs.500 to **A/c-2**, but after performing debit operation somehow system goes down or power failure happened and credit operation not complete, because of that we loss Rs.500. This leads database towards **in-consistence state**.
- It is difficult to ensure atomicity in a file system, while in DBMS you do not need to worry about it.

7) Concurrent Access Anomalies

- In any system, for better performance and faster response, multiple users are allowed to access data concurrently.

- Assume that **A/c-1** having balance Rs.5000, now consider concurrent access of **A/c-1**. A person access A/c-1 and debit Rs.1000 by reading A/c-1 data from database, but before writing change back to database, simultaneously someone performed debit operation on same account worth Rs.500. So, it will read Rs.5000 as balance and after debit Rs.500 remaining balance is Rs. 4500. But after two simultaneous transaction worth Rs.1000 and Rs.500 actual balance would be Rs.3500.
- In file system, concurrent data access should be performed under some supervision, which is usually some program.
- Due to lack of coordination between different application program, it is difficult to achieve this in file system.

8) Security Problems

- Database should be accessible in a limited way to different users.
- Each user is allowed to access only those data, which is required for his/her application.
- In file processing system, different programmers are adding their application program; it is difficult to achieve security constraints.
- Example: Only a customer can check his/her account detail, no one else have to rights to access his/her account.

Advantage of using DBMS over the traditional file system

- **Fast Data Access:** The data response time increases in DBMS. **Example:** If we search for any train in IRCTC then we get results about that train only in a fraction of seconds. But the file system would have given us the entire file. Such a file size is large hence its access time will also be more.
- **Minimized Data Redundancy:** DBMS has different constraints like primary key constraints which helps in ensuring that we store only unique data in our database i.e. same data can't be stored in more than one places.